Homework 5

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1. Let (X, d_X) be a metric space, and denote by $C_b(X)$ the set of all bounded continuous functions $f : X \to \mathbb{R}$ where $\mathbb{R}$ is equipped with the Euclidean metric. Define a metric on $C_b(X)$ as

$$d(f, g) = \sup_{x \in X} |f(x) - g(x)|.$$

(you do not need to show that this is a metric.) Show that $C_b(X)$ is complete. Use this to deduce that $C(X)$, the set of all continuous functions on X with the same metric, is complete as long as X is compact.
Hint: If (f_n) is Cauchy w.r.t. the metric defined above, then for each $x \in X$, $(f_n(x))$ is a Cauchy sequence in $\mathbb{R}$. This shows that $(f_n(x))$ converges to a limit, which we can denote by $f(x)$. Then prove that $f : X \to \mathbb{R}$ is bounded and continuous.

Proof. As noted above, if (f_n) is Cauchy under the sup metric, $(f_n(x))$ is Cauchy for all $x \in X$. Since $(f_n(x))$ is a Cauchy sequence in $\mathbb{R}$, it has a unique limit in $\mathbb{R}$. Denote this limit as $f(x)$ for each x. The key is to notice that $(f_n(x))$ converges uniformly. Indeed, given any $y \in X$ and $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that

$$\delta > \sup_{x \in X} |f(x) - f_n(x)| \geq |f(y) - f_n(y)|.$$

Since the choice of y was arbitrary, we get $f_n \to f$ uniformly. Now since the uniform limit of bounded functions is bounded and the uniform limit of continuous functions is continuous, we have $f \in C_b(X)$.

Finally, the extreme value theorem tells us that if X is compact, any $f \in C(X)$ is bounded and is thus in $C_b(X)$, so any Cauchy sequence in $C(X)$ is a Cauchy sequence in $C_b(X)$, and thus converges to a limit in $C_b(X) \subset C(X)$. □

2. Find a counterexample for each of the following:

(a) The conclusion of Question 1 fails if $C_b(X)$ is replaced by $C(X)$ without assuming that X is compact.

(b) The conclusion of Question 1 fails if $\mathbb{R}$ in the definition of $C_b(X)$ is replaced by $\mathbb{Q}$ (or by any subset of $\mathbb{R}$ that is not closed).

Proof. (a) The sup metric isn't even a metric on $C(X)$ if X is not compact, since for instance the "distance" between the constant function $\mathbf{0}$ and the function $y = x$ on $C(\mathbb{R})$ would be ∞, which is not an allowed value for a metric.

(b) Take $X = \mathbb{R}$, and let (f_n) be a sequence of rational constant functions whose values converge to π. Then each $f_n \in C(\mathbb{R})$ and (f_n) is Cauchy, but it does not converge to a limit in $C(\mathbb{R})$ (under this definition).

□

3. On $C_b(\mathbb{R})$, define

$$T(f)(x) = \frac{1}{2}f(x+1).$$

Show that T is a contraction, and find its unique fixed point.

Proof. Let us compare two functions $f, g \in C_b(\mathbb{R})$. We have

$$\begin{aligned} d(T(f), T(g)) &= \sup_{x\in X} |T(f)(x) - T(g)(x)| \\ &= \sup_{x\in X} \left|\frac{1}{2}f(x+1) - \frac{1}{2}g(x+1)\right| \\ &= \sup_{x\in X} \frac{1}{2} |f(x+1) - g(x+1)| \\ &= \frac{1}{2} \sup_{x\in X} |(f-g)(x+1)| \\ &= \frac{1}{2} \sup_{x\in X} |(f-g)(x)| \\ &= \frac{1}{2} \sup_{x\in X} |f(x) - g(x)| \\ &= \frac{1}{2} d(f,g). \end{aligned}$$

We haven't needed to introduce an inequality (though we could) since we only have the "equals" part of the condition for a contraction, but we do indeed have a contraction with constant $1/2$. Since $f = \mathbf{0}$ is clearly *a* fixed point, it must be the *unique* fixed point. □

4. Show that the integral equation

$$f(x) = \int_0^x \frac{1}{2} f(t)dt + g(x).$$

has a unique solution on $C([0,1])$.

Proof. Define a function $T : C([0,1]) \to C([0,1])$ by

$$T(f)(x) = \int_0^x \frac{1}{2} f(t)dt + g(x).$$

This is well defined since every continuous function is Riemann integrable. The codomain is correct as written since the integral of a continuous function is continuous, and the sum of continuous functions is continuous. Given $f, h \in C([0,1])$, we have

$$\begin{aligned}
d(T(f), T(h)) &= \sup_{x\in[0,1]} \left| \left(\int_0^x \frac{1}{2} f(t)dt + g(x) \right) - \left(\int_0^x \frac{1}{2} h(t)dt + g(x) \right) \right| \\
&= \sup_{x\in[0,1]} \left| \frac{1}{2} \int_0^x f(t) - h(t)dt \right| \\
&= \frac{1}{2} \sup_{x\in[0,1]} \left| \int_0^x f(t) - h(t)dt \right| \\
&\leq \frac{1}{2} \sup_{x\in[0,1]} \int_0^x |f(t) - h(t)|\, dt \\
&\leq \frac{1}{2} \sup_{x\in[0,1]} \int_0^x d(f,h)dt \\
&= \frac{1}{2} \sup_{x\in[0,1]} x d(f,h) \\
&= \frac{1}{2} d(f,h).
\end{aligned}$$

Thus T is a contraction with constant $1/2$, so it must have a unique fixed point, which is a solution to our integral equation. □